

Phase-1 model families on Craftax-Classic: pure RL vs Transformer vs world-model features vs ChunkPPO

Run 2026-09-30/10-01, local RTX 3060 Ti; branch `claude/reasoning-model-baselines-38c2e9`

Summary

- **None of the four requested families beat pure RL** (100k and 1M steps): pure Transformer, Transformer + Idea-6 world-model (WM) features, ChunkPPO ($k=1$).
- **Trained WM features = frozen random WM**: reward_pct -0.05 ± 0.36 (100k), $+1.4 \pm 1.2$ (1M); both n.s.
- **One derivative model wins at 1M**: `tf_gru` = tile-token Transformer encoder + GRU memory (PPO, 0.45M params): 39.2 ± 1.3 / 11.5 ± 0.5 vs best pure RL (GRU) 33.6 ± 0.8 / 9.1 ± 0.5 , $\Delta = +5.6 \pm 1.5$ / $+2.3 \pm 0.7$. Neither part alone does it (Transformer without memory 29.2; GRU without Transformer 33.6; CNN+GRU control 33.0). No WM, search or chunking involved. Designed *after* seeing the other 1M results; passed one clean test on 10 fresh seeds.
- **No roadmap gate claimed**. G1.0 fails: best baseline 33.6/9.1 vs required ≥ 42.7 / ≥ 9.6 ($0.9\times$ MFRL_4M_reimpl, Dedieu 2025 Table 1), so Phase-1 comparisons are internal only. `tf_gru` reward $39.2 < 47.40$; score $11.45 - 2 \cdot 0.45 = 10.55 < 10.71$.
- 100k runs (requested budget) show no separation between architectures. The 1M runs are a deliberate extension (cost 5–60 min/seed locally; only they can speak to the gates). Nothing above 1M was run.

Setup

Env/obs: `CraftaxClassicSymbolicEnv`, default params (limit 10 000 steps, no T cap), raw 1345-d symbolic obs ($7\times 9\times 21$ tiles + 22 scalars), 17 actions, unchanged reward. **Protocol**: *EP-A-mini* (≤ 100 k steps, 98 304–99 840 executed; never gating) and *EP-A* (999 424 steps, 10 seeds 0–9). Sampled policy, first episode of 256 fresh envs/seed, final params; tuning seeds ≥ 1000 (disjoint), same grid for every arm (100k: $n16\times T32$ / $n64\times T64 \times \text{lr}\{3e-4, 1e-3, 3e-3\}$, 3 seeds; 1M: $\text{lr}\{3e-4, 1e-3, 3e-3\}$ at $n64\times T64$, 2 seeds). Selection = mean reward_pct on tuning seeds. **Metrics** (*metric_check* done by code reading + unit tests): reward_pct = mean (distinct achievements/22) $\times 100$ via `masked_achievements()`; score_pct = $\exp(\frac{1}{22} \sum_i \ln(1 + s_i)) - 1$. Untrained init policy: 10.17 ± 0.15 / 1.75 ± 0.03 .

Results

Mean $\pm$ SE over 10 seeds; Δ = difference to the reference arm $\pm$ Welch SE. Per-seed values: `output/phase1/final*_all/`.

arm	description	params_total	deployed
<code>ppo_mlp</code>	PPO, two tanh MLPs 3×512 (pure RL)	2.44M	1.22M
<code>ppo_gru</code>	embed $\rightarrow$ GRU(256) $\rightarrow$ heads (pure RL, recurrent)	0.88M	0.81M
<code>tf</code>	Transformer d64/2L over 63 tiles + scalars + 4 history steps + 17 candidate-action tokens + CLS; logits from candidate tokens	0.20M	0.20M
<code>tf_wm</code>	<code>tf</code> + action-conditioned 1-step WM features per candidate action (no noop future; trained only on the executed transition; stop-grad)	0.98M	0.98M
<code>tf_wm_random</code>	control: same, WM frozen at random init	0.98M	0.98M
<code>chunkppo_k1</code>	<code>chunk_ppo.py</code> , chunk Transformer actor over [state, 4 hist], MLP critic, $k=1$	1.41M	0.19M
<code>tf_gru</code>	per-step <code>tf</code> encoder (no history tokens) $\rightarrow$ GRU(256)	0.45M	0.45M
<code>cnn_gru</code>	control: 2-layer CNN encoder $\rightarrow$ same GRU	1.59M	1.53M

Table 1: *
Italic = derivative arms added after the main comparison.

$\leq 100k$ steps (EP-A-mini), reference = <code>ppo_mlp</code>				
arm	reward_pct	score_pct	Δ reward	Δ score
<code>ppo_mlp</code>	20.10 ± 0.16	3.71 ± 0.09	–	–
<code>ppo_gru</code>	20.06 ± 0.28	3.16 ± 0.10	-0.03 ± 0.32	-0.56 ± 0.14
<code>cnn_gru</code>	18.97 ± 0.28	3.00 ± 0.09	-1.13 ± 0.33	-0.71 ± 0.13
<code>chunkppo_k1</code>	18.60 ± 0.33	3.03 ± 0.12	-1.50 ± 0.36	-0.68 ± 0.16
<code>tf</code>	18.17 ± 0.28	3.31 ± 0.07	-1.92 ± 0.32	-0.41 ± 0.12
<code>tf_gru</code>	17.94 ± 0.16	3.33 ± 0.09	-2.15 ± 0.23	-0.38 ± 0.13
<code>tf_wm_random</code>	17.69 ± 0.31	2.92 ± 0.08	-2.40 ± 0.35	-0.80 ± 0.13
<code>tf_wm</code>	17.64 ± 0.18	3.07 ± 0.06	-2.46 ± 0.24	-0.65 ± 0.11

Table 2: *
Derivative arms use the config selected at 1M (no 100k tuning). 100k is too little for any architecture to leave the starter achievements (nothing reaches stone).

Exploratory variants (100k unless noted; tuning seeds, reduced grid, *not* equal effort)

variant	best reward_pct	note
<code>tf</code> / <code>ppo_mlp</code> reference (full grid)	18.9 / 20.7	
<code>tf</code> + separate 3×512 critic	18.2	shared value head is not the cause
<code>tf</code> , no history tokens	18.7	history window irrelevant
<code>tf</code> + aux loss on executed candidate token (0.1 / 1.0)	19.1 / 19.1	$\leq +0.2$, below MLP
<code>tf</code> d128, 3 layers	19.2	+0.3
<code>tf_wm</code> + imagined 1-step Q feature / random-WM control	18.1 / 18.7	control $\geq$ trained
GRU improvement attempts (1M): ent 0.003 / γ	34.1 / 31.9 / 30.6 / 30.6 / 27.2	none near G1.0
0.995 / λ 0.95 / ent 0.02 / 8 epochs		
<code>tf_gru</code> + WM features, trained / random (1M, 1 seed, interrupted)	31.5 / 33.1	vs 41.4 without, same seed

Verdicts

- Pure RL:** the GRU is the suitable baseline (33.6/9.1 at 1M) but *not* calibrated to the paper’s MFRL (G1.0 fails). Untested likely causes: image obs + CNN, larger nets, longer tuning in the references.
- Pure Transformer:** 100k -1.9 ± 0.3 vs MLP; 1M -0.9 ± 0.6 vs MLP (n.s.) with $12\times$ fewer params, but -4.3 ± 0.9 vs GRU. A 4-step history window is no substitute for recurrent memory.

1M steps (EP-A protocol), reference = ppo_gru				
arm	reward_pct	score_pct	Δ reward	Δ score
tf_gru	39.19 $\pm$ 1.27	11.45 $\pm$ 0.45	+5.61 $\pm$ 1.52	+2.33 $\pm$ 0.66
ppo_gru	33.57 $\pm$ 0.83	9.12 $\pm$ 0.48	–	–
cnn_gru	32.95 $\pm$ 0.57	8.35 $\pm$ 0.40	–0.63 $\pm$ 1.00	–0.77 $\pm$ 0.63
ppo_mlp	30.14 $\pm$ 0.46	7.08 $\pm$ 0.36	–3.43 $\pm$ 0.95	–2.04 $\pm$ 0.60
tf	29.24 $\pm$ 0.36	6.67 $\pm$ 0.31	–4.34 $\pm$ 0.91	–2.46 $\pm$ 0.57
tf_wm	28.58 $\pm$ 0.75	6.71 $\pm$ 0.47	–5.00 $\pm$ 1.12	–2.41 $\pm$ 0.67
tf_wm_random	27.16 $\pm$ 0.93	5.86 $\pm$ 0.51	–6.42 $\pm$ 1.25	–3.26 $\pm$ 0.71
chunkppo_k1	23.90 $\pm$ 0.34	4.58 $\pm$ 0.12	–9.68 $\pm$ 0.90	–4.54 $\pm$ 0.50

Table 3: *

Selected lr: 1e-3 for all except **tf_wm** (3e-3) and **chunkppo_k1** (3e-4); all n64 $\times$ T64. **tf_gru** tuning lr: 3e-4 $\rightarrow$ 27.3, 1e-3 $\rightarrow$ 42.3, 3e-3 $\rightarrow$ 30.0 (1 seed). Episodes are short for every arm (mean 140–230 steps, none censored), so these are early-game results.

unlock % (1M)	tf_gru	gru	cnn_gru	mlp	tf	tf_wm	wm_rnd	chunk
collect_wood	99.1	99.3	98.6	97.8	97.6	97.0	95.9	89.3
place_table	95.1	94.7	93.2	88.7	90.9	87.3	83.9	69.4
make_wood_pickaxe	74.8	56.2	58.4	48.8	48.2	33.1	31.6	20.2
make_wood_sword	67.5	53.2	58.8	44.8	35.9	32.3	29.2	20.8
collect_stone	54.7	30.4	25.8	8.3	11.4	11.5	8.2	1.3
place_stone	44.8	17.1	8.8	3.5	5.0	3.9	3.5	0.5
place_furnace	49.3	23.8	9.5	3.1	6.1	3.8	3.2	0.4
defeat_zombie	29.1	13.5	11.5	19.7	11.7	13.6	11.3	7.7
eat_cow	17.3	14.1	13.0	37.0	11.2	13.4	6.5	7.3

Table 4: *

Only **tf_gru** reaches the stone/furnace tier regularly. (collect_sapling, place_plant, wake_up: $\approx$ 70–98% for all arms.)

3. **WM features (Idea 6, step-by-step)**: no benefit; consistent with the prior objection that a deterministic function of the same observation adds no information. Imagined-Q and aux-loss variants do not change this.
4. **ChunkPPO $k=1$** : worst at 1M. At $k=1$ it is PPO with a single-state-token Transformer actor; the tile-token encoder is 5 points better, so the weakness is the actor/critic design, not the inactive chunk machinery. $k > 1$ untested (Phase 3 locked).
5. **tf_gru**: the only clear gain; appears only at 1M (not at 100k). Treat as a new hypothesis.

Caveats and next steps

Symbolic obs only (papers: image + CNN); small equal tuning (3 lr points at 1M; architecture sizes untuned; GRU baseline received extra single-change variants); derivative arms added post hoc; sampled policy only. Result JSONs record HEAD at launch, while some derivative-arm code was still uncommitted then (final code is in the commit containing this note).

Next: (1) calibrate the baseline vs the paper with image obs + CNN/Impala + GRU at equal tuning (needed before any gate); (2) if WM is revisited, use it for data reuse (Dyna-style imagined updates), not as features; (3) tune **tf_gru** with the same effort as the GRU baseline and add an image variant.

Incident note (GPU slowdown, restart 2026-10-01 12:40)

Most likely GPU/VRAM oversubscription by my jobs (single GPU shared with the display; XLA memory caps of concurrent containers summed above free VRAM; 82–100% utilisation for $\sim$ 18 h; parallel jobs gave no speed-up) rather than RAM or a leaked process. Supporting: 12 `nvlddmkm` Event-13 driver errors (the first ones coincide with containers failing at start with “no supported devices” CUDA errors). Mitigation: one job at a time, total memory fraction ≤ 0.6 , check the driver, keep the repo off the RAM disk. Not proven.

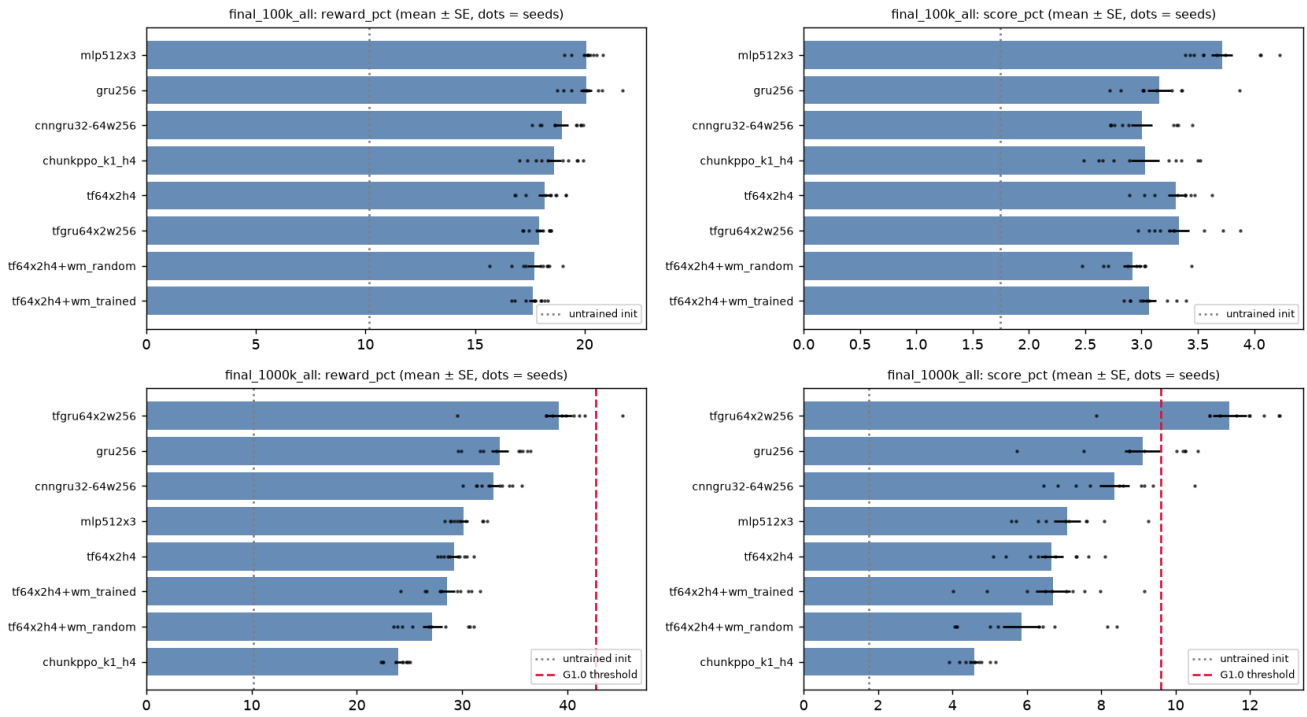


Figure 1: reward_pct and score_pct per arm (bars = mean $\pm$ SE, dots = seeds). Top: $\leq 100k$; bottom: 1M with the G1.0 thresholds (red dashed) and the untrained policy (grey dotted).